

Experiment -1

Linux Programming Environment & Basic Commands

A) Setting up of Linux Programming Environment

We can do C programming online or offline according to the needs. It is required for us to keep working on a Linux Environment.

Online:

Cocalc.ai – Cocalc.ai provides an online environment for programming

There are few steps we have to follow for it-

1. Open cocalc.ai then open a project
2. Open the programming environment
3. Create c file
4. Compile and execute the file

Offline:

Methods: Dual boot, Single boot, Bootable Pendrive, VM

1. DUAL BOOT- Both linux and other operating system can be installed in same computer .But we have to select one system that we have to operate .
2. SINGLE BOOT- In computer linux is installed as a one main operating system
3. BOOTABLE PENDRIVE – Using a bootable USB pendrive linux can be started.
4. VIRTUAL MACHINE (VM) – Linux is run as a virtual computer inside an another operating system.

1. Virtual Machine Manager (Hypervisor)

Type-1 (Baremetal)

A type-1 hypervisor can be run directly on the physical hardware.

E.g. Hyper-V /VMware ESXi

Type-2 (Hosted)

A type-2 hypervisor can run on top of an existing operating system .

E.g Oracle Virtualbox, VM Ware.

We would be using Oracle Virtualbox.

2. Installation of VMM & VM

Step-1: Download Oracle Virtual Box for “Windows hosts”

Step-2: Download Pre-built Kali Linux VM for Virtualbox

Step-3: Extracted the Kali Linux downloaded file

Step-4: Install Virtualbox

Step-5: Open Virtual box

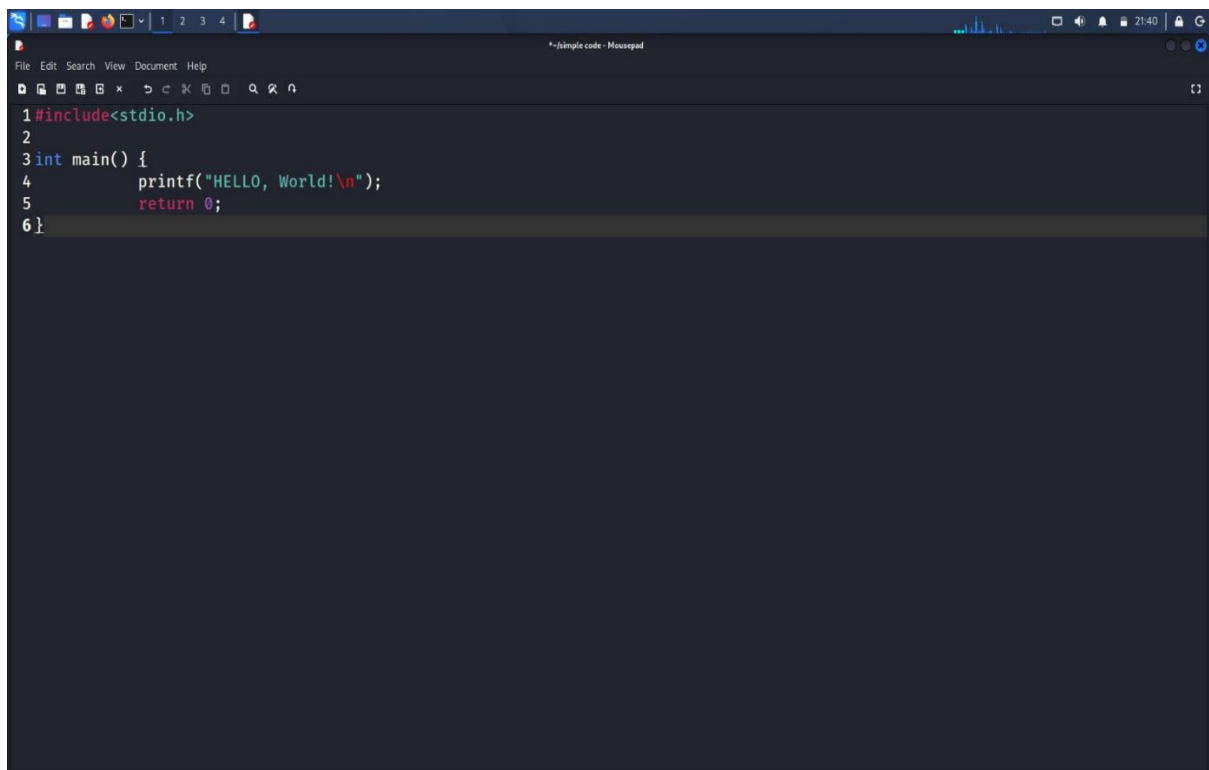
Step-6: Open the “saved VM”

Step-7: Start the VM

Thus, the installation of VMM (Hypervisor Type-2 Oracle VirtualBox) & VM (Kali Linux) is completed.

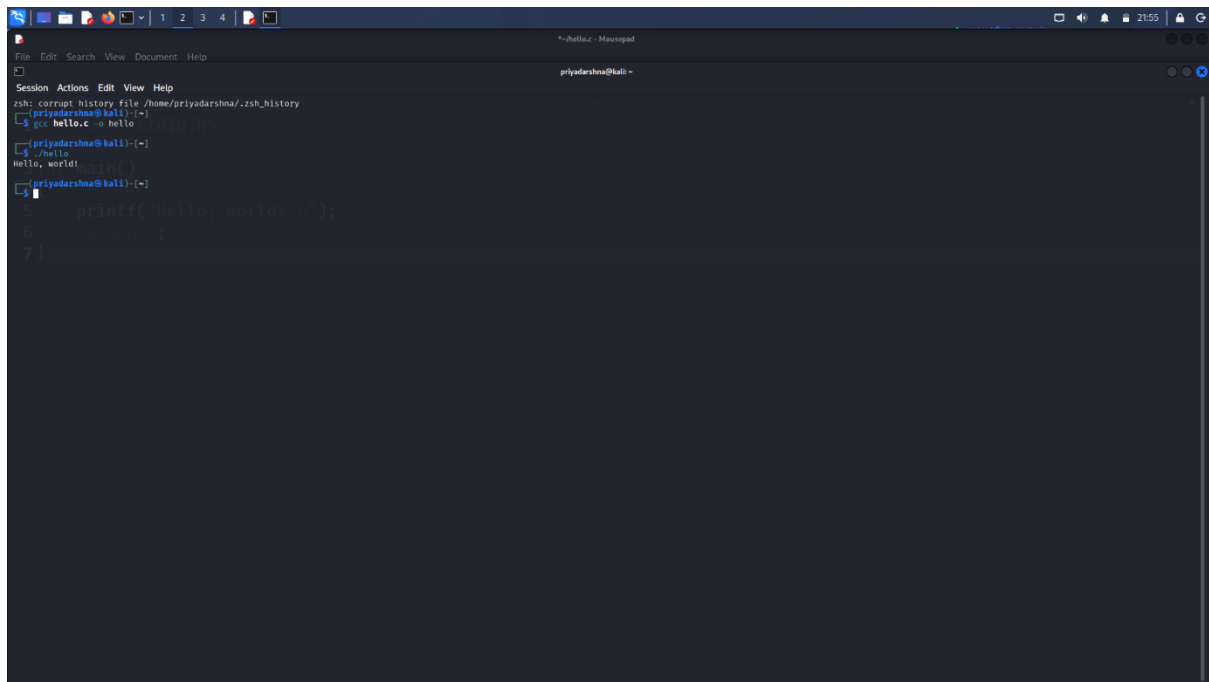
Step-8: Open terminal and create the following simple program, compile and execute to understand the working of the environment.

Implementation and Execution of a simple C programming in kali linux –

A screenshot of a code editor window titled "simple code - Mousepad". The editor displays a simple C program with the following code:

```
1 #include<stdio.h>
2
3 int main() {
4     printf("HELLO, World!\n");
5     return 0;
6 }
```

The code is written in a dark-themed editor with syntax highlighting. The file explorer on the left shows a single file named "simple.c". The top status bar indicates the file is "simple code - Mousepad".



```
File Edit Search View Document Help
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
~(priyadarshna@kali)~
$ gcc hello.c -o hello
~(priyadarshna@kali)~
$ ./hello
hello, world:
~(priyadarshna@kali)~
$
```

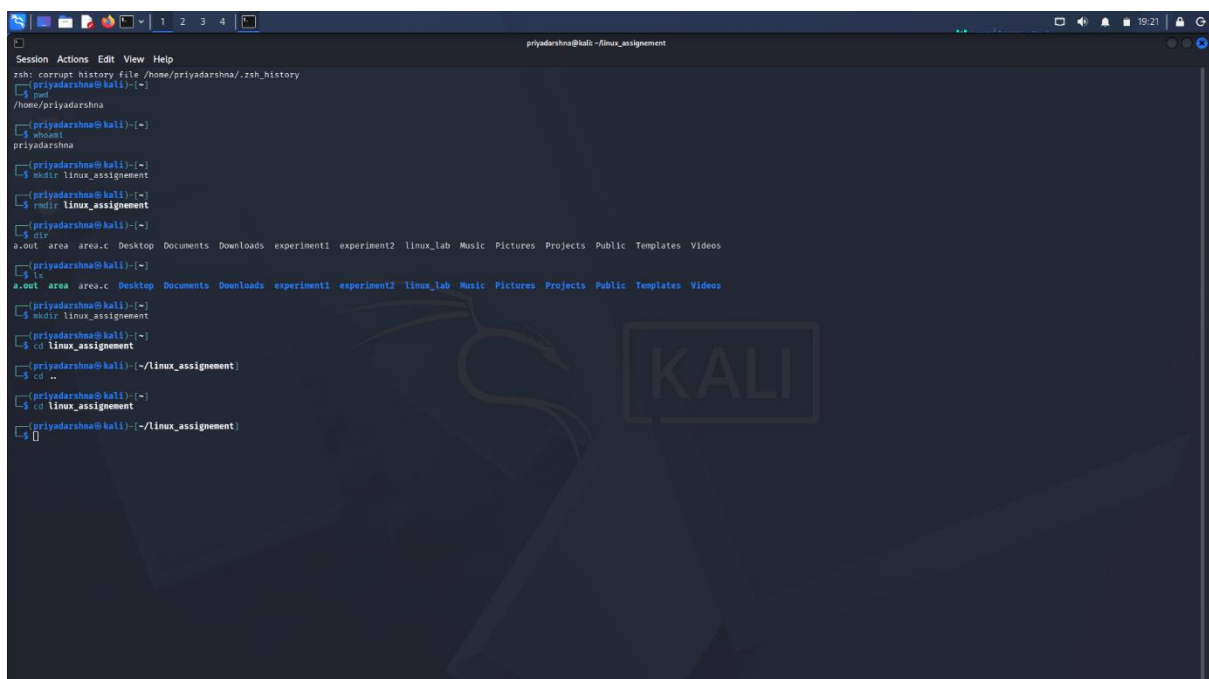
The screenshot shows a terminal window with a dark background. The top bar includes a menu (File, Edit, Search, View, Document, Help) and a session title 'priyadarshna@kali'. The terminal content shows a user at the prompt '~(priyadarshna@kali)~' running the command '\$ gcc hello.c -o hello'. The next prompt shows the user running '\$./hello', which outputs 'hello, world:'. The prompt returns to '~(priyadarshna@kali)~'.

B) Working with Linux Commands

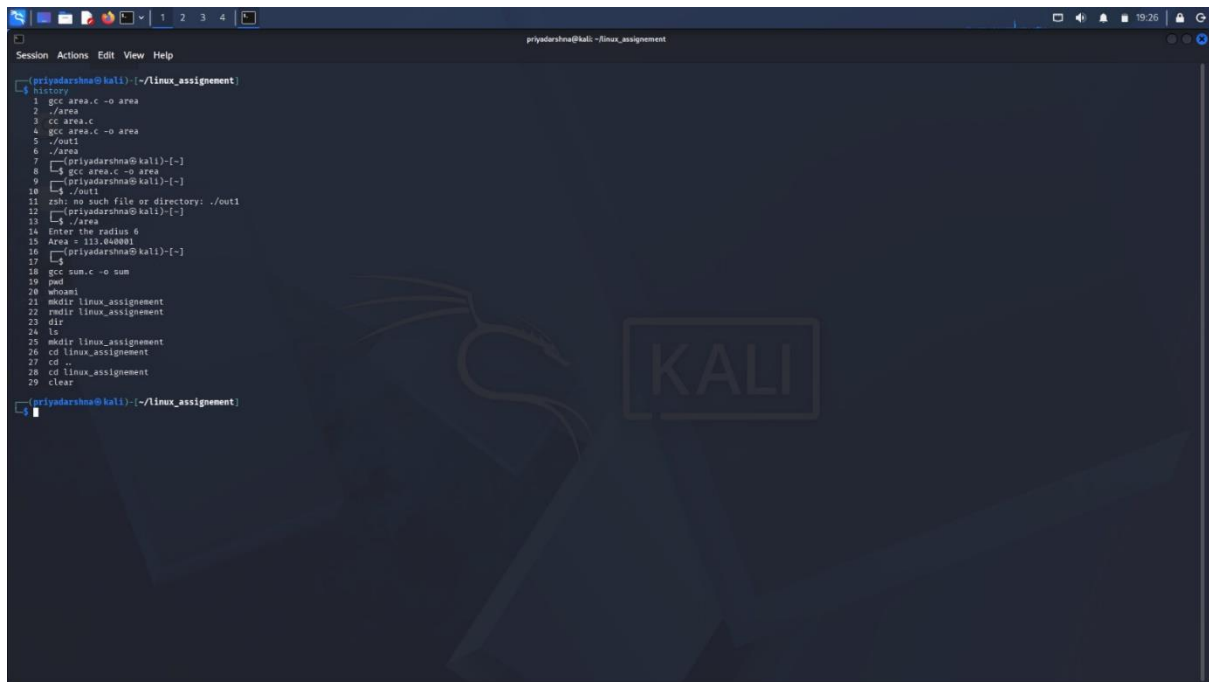
Execute the following commands in your environment, take output snippets, and provide a brief explanation.

- Terminal basics: pwd, mkdir, rmdir, dir, ls, cd, cd .., clear, history, man, whoami
- pwd – Shows the current directory.
- mkdir – Creates a directory named.
- dir – Lists files and directories.
- ls - Lists files and directories.

- `cd ..` – Returns to the parent directory.
- `rmdir` – Deletes the empty test directory.
- `Clear` – Cleans the terminal screen.
- `History` – Shows previously executed commands.
- `man ls` – opens the manual for `ls`; press `q` to exit.
- `Whoami` – Displays the current username.



```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
~ (priyadarshna@kali) ~
$ pwd
/home/priyadarshna
~ (priyadarshna@kali) ~
$ whoami
priyadarshna
~ (priyadarshna@kali) ~
$ mkdir linux_assignment
~ (priyadarshna@kali) ~
$ rmdir linux_assignment
~ (priyadarshna@kali) ~
$ dir
a.out area area.c Desktop Documents Downloads experiment1 experiment2 linux_lab Music Pictures Projects Public Templates Videos
~ (priyadarshna@kali) ~
$ ls
a.out area area.c Desktop Documents Downloads experiment1 experiment2 linux_lab Music Pictures Projects Public Templates Videos
~ (priyadarshna@kali) ~
$ mkdir linux_assignment
~ (priyadarshna@kali) ~
$ cd linux_assignment
~ (priyadarshna@kali) ~/linux_assignment
$ cd ..
~ (priyadarshna@kali) ~
$ cd linux_assignment
~ (priyadarshna@kali) ~/linux_assignment
$
```

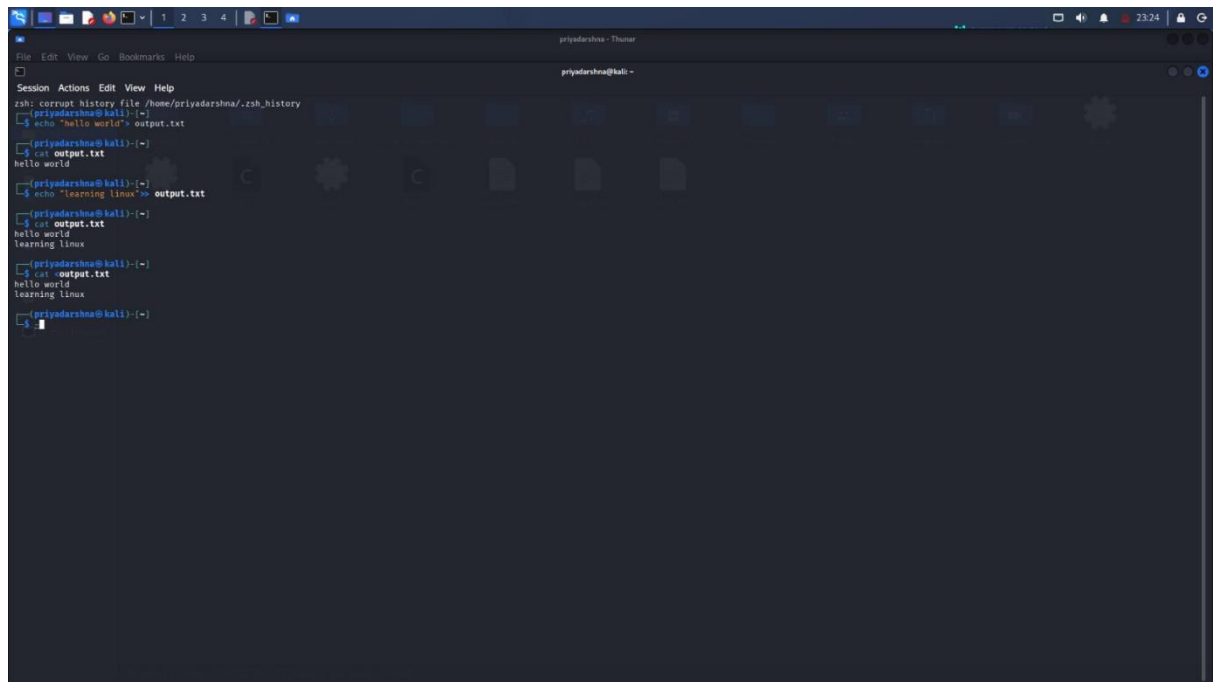


```
Session Actions Edit View Help
priyadarshna@kali: ~/linux_assignment

priyadarshna@kali:~/linux_assignment$ history
1 gcc area.c -o area
2 ./area
3 cc area.c
4 gcc area.c -o area
5 ./out1
6 ./area
7
8 priyadarshna@kali:~/linux_assignment$ gcc area.c -o area
9 priyadarshna@kali:~/linux_assignment$ ./out1
10
11 rm: no such file or directory: ./out1
12 priyadarshna@kali:~/linux_assignment$ ./area
13
14 Enter the radius 6
15 Area = 113.040001
16 priyadarshna@kali:~/linux_assignment$
17 gcc sum.c -o sum
18
19 ped
20 whoami
21
22 mkdir linux_assignment
23
24 ls
25
26 cd linux_assignment
27
28 cd linux_assignment
29 clear
30
```

*Files: vi,file,rm,cp,mv,cat,more,less,head,tail,chmod

- vi- Opens a text file for creating or editing.
- file- Displays the type of a file.
- rm - Removes a file.
- cp - Copies a file or directory.
- mv- Moves or renames a file.
- cat – Displays the complete contents of a file.
- more- Displays file contents one screen at a time.
- less- Displays file contents with forward and backward navigation.
- head- Shows the first 10 lines of a file.
- tail- Shows the last 10 lines a file.
- chmod- Changes file permission.



```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ echo "hello world"> output.txt
[priyadarshna@kali:~]$ cat output.txt
hello world
[priyadarshna@kali:~]$ echo "learning linux"> output.txt
[priyadarshna@kali:~]$ cat output.txt
hello world
learning linux
[priyadarshna@kali:~]$ cat output.txt
hello world
learning linux
[priyadarshna@kali:~]$
```

NAME – PRIYADARSHNA

BATCH – 15

SAPID – 590032928

TOPIC – LINUX PROGRAMMING ENVIRONMENT & BASIC COMMANDS

